

Characterize the max-min revenue-optimal auction for n at least three i.i.d. bidders

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Distributional Robustness: From Pricing to Auctions*.

1 The open problem

This paper addresses an open problem posed by Nir Bachrach, Inbal Talgam-Cohen in *Distributional Robustness: From Pricing to Auctions* [1] (Economics and Computation (EC), 2022), Section 5 (Discussion), page 29. The website catalogue entry is problem 20 (W4280639630_p0) of the [Open Problems in Operations Research](#) collection.

Determine (or provide a tight quantitative characterization of) an equilibrium (M^, F^*) of the general max-min revenue game for $n \geq 3$ i.i.d. bidders with $F \in \mathcal{F}(\mu, \bar{v})$, i.e., solve*

$$\max_{M \text{ truthful}} \min_{F \in \mathcal{F}(\mu, \bar{v})} \text{Rev}_F(M)$$

by identifying an optimal truthful mechanism M^ and the corresponding worst-case distribution F^* (and hence the max-min value $\min_{F \in \mathcal{F}(\mu, \bar{v})} \text{Rev}_F(M^*)$).*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

This repair yields rigorous partial progress, not a complete solution of the unrestricted max-min problem. In particular, the adversary-first value below remains only an upper bound on the desired max-min value.

Throughout, normalize the value bound to 1 and write

$$\mathcal{F}_m = \{F \text{ on } [0, 1] : \mathbb{E}_F V = m\},$$

$$V_n(m) = \sup_M \inf_{F \in \mathcal{F}_m} \text{Rev}_F(M), \quad D_n(m) = \inf_{F \in \mathcal{F}_m} \text{OPT}_n(F).$$

We always have $V_n(m) \leq D_n(m)$.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_071015_422197c1, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

2 Exact adversary-first value

Define

$$A(t) = t(1 - \log t), \quad K_n(t) = 1 - (1 - t)^n.$$

Let $\tau_n \in (0, 1)$ be the unique solution of

$$-\log \tau_n = \frac{n(1 - \tau_n)}{1 + (n - 1)\tau_n},$$

and set

$$m_n = A(\tau_n), \quad \kappa_n = \frac{(1 - \tau_n)^n}{1 - m_n} = (1 + (n - 1)\tau_n)(1 - \tau_n)^{n-2}.$$

For $m \in [0, 1]$, let $t_m = A^{-1}(m)$. Then

$$D_n(m) = U_n(m) = \begin{cases} K_n(t_m), & 0 \leq m \leq m_n, \\ 1 - \kappa_n(1 - m), & m_n \leq m \leq 1. \end{cases}$$

The minimizing law is unique.

For $0 < m \leq m_n$, its upper quantile is

$$v_m(q) = \begin{cases} 1, & q \leq t_m, \\ t_m/q, & q > t_m, \end{cases}$$

or, equivalently,

$$F_m^\dagger(x) = \begin{cases} 0, & x < t_m, \\ 1 - t_m/x, & t_m \leq x < 1, \end{cases}$$

with an atom of mass t_m at 1.

For $m_n \leq m < 1$, put

$$\lambda = \frac{1 - m}{1 - m_n}, \quad a = 1 - \lambda.$$

Then

$$v_m(q) = \begin{cases} 1, & q \leq \tau_n, \\ a + \lambda\tau_n/q, & q > \tau_n. \end{cases}$$

Its continuous support begins at $a + \lambda\tau_n$, where its tail is

$$\Pr(V \geq x) = \frac{\lambda\tau_n}{x - a},$$

and it has an atom of mass τ_n at 1. The boundary cases $m = 0$ and $m = 1$ correspond to $F^\dagger = \delta_0$ and $F^\dagger = \delta_1$, respectively.

Proof. Let $v_F(q)$ be a randomized upper quantile, let $R_F(q) = qv_F(q)$, and let H_F be the least concave nondecreasing majorant of R_F . Atomic Myerson ironing gives

$$\text{OPT}_n(F) = n \int_0^1 H'_F(q)(1 - q)^{n-1} dq = n(n - 1) \int_0^1 H_F(q)(1 - q)^{n-2} dq. \quad (1)$$

Here H'_F is interpreted almost everywhere, and randomized tie-breaking is used inside atoms.

Because $H_F \geq R_F$,

$$I_F = \int_0^1 \frac{H_F(q)}{q} dq \geq m. \quad (2)$$

Conversely, if

$$H(0) = 0, \quad H \text{ concave and nondecreasing,} \quad 0 \leq H(q) \leq q,$$

then $H(q)/q$ is a decreasing quantile in $[0, 1]$. If $I_F > m > 0$, replacing H_F by $(m/I_F)H_F$ produces a mean- m law and strictly reduces (1). Hence

$$D_n(m) = \min_{\substack{H(0)=0, \ H \text{ concave nondecreasing} \\ H \leq q, \ \int H(q)/q dq = m}} n(n-1) \int_0^1 H(q)(1-q)^{n-2} dq. \quad (3)$$

Every admissible H has a unique probability representation

$$H(q) = \int_{[0,1]} \min\{q, t\} \nu(dt). \quad (4)$$

Indeed, $S(q) = H'_+(q)$ is nonincreasing and satisfies $0 \leq S(q) \leq 1$. Define

$$\nu((q, 1]) = S(q), \quad \nu(\{0\}) = 1 - H'_+(0).$$

Then

$$H(q) = \int_0^q S(u) du = \int \min\{q, t\} \nu(dt).$$

The tail values $S(q)$ determine ν on $(0, 1]$, while probability normalization determines the otherwise invisible mass at 0, proving uniqueness.

Tonelli's theorem now yields

$$\int_0^1 \frac{H(q)}{q} dq = \int A(t) \nu(dt)$$

and

$$n(n-1) \int_0^1 H(q)(1-q)^{n-2} dq = \int K_n(t) \nu(dt). \quad (5)$$

Thus $D_n(m)$ is the lower convex envelope of the function ϕ defined by

$$\phi(A(t)) = K_n(t).$$

It remains to identify this envelope globally. Put $L = -\log t$ and

$$s(t) = \phi'(A(t)) = \frac{n(1-t)^{n-1}}{L}.$$

Then

$$\frac{s'(t)}{s(t)} = \frac{1-t + (n-1)t \log t}{tL(1-t)}. \quad (6)$$

The numerator changes sign exactly once: it is positive precisely when

$$\frac{1-t}{t(-\log t)} > n-1,$$

and the left-hand side decreases strictly from $+\infty$ to 1. Hence ϕ is strictly convex and then strictly concave.

Define the endpoint-tangent gap

$$h(t) = 1 - K_n(t) - s(t)(1 - A(t)).$$

A cancellation gives

$$h'(t) = -(1 - A(t))s'(t). \quad (7)$$

Moreover $h(0+) = 1$, while

$$h(t) = (1-t)^n \left[1 - \frac{n(1-A(t))}{(1-t)(-\log t)} \right]$$

and

$$\frac{1-A(t)}{(1-t)(-\log t)} \longrightarrow \frac{1}{2} \quad (t \uparrow 1).$$

Since $n \geq 3$, we have $h(t) < 0$ near 1. Equation (7) therefore implies that h has exactly one zero τ_n , occurring strictly before the inflection point. Its zero equation is precisely

$$-\log \tau_n = \frac{n(1-\tau_n)}{1+(n-1)\tau_n}.$$

Let

$$\ell(x) = K_n(\tau_n) + s(\tau_n)(x - A(\tau_n))$$

be this tangent; it passes through $(1, 1)$. To rule out additional contacts, put

$$d(t) = K_n(t) - \ell(A(t)).$$

Then

$$d(\tau_n) = d(1) = 0, \quad d'(t) = (-\log t)(s(t) - s(\tau_n)). \quad (8)$$

Unimodality of s shows that d decreases to 0 on $(0, \tau_n)$, then increases, and finally decreases to 0 at 1. Consequently,

$$d(t) > 0 \quad (t \notin \{\tau_n, 1\}). \quad (9)$$

Thus the lower convex envelope equals ϕ up to $A(\tau_n)$, followed by the tangent chord to $(1, 1)$, with no hidden contacts.

Strict convexity and (9) also give uniqueness: below m_n , $\nu = \delta_{t_m}$; above m_n ,

$$\nu = \lambda \delta_{\tau_n} + (1-\lambda) \delta_1.$$

Equality in (2) then forces $R_F = H_F$ almost everywhere, so the stated laws are the unique minimizing distributions. $\square$

For $n = 3$,

$$\tau_3 \approx 0.116586033, \quad m_3 \approx 0.367144084, \quad \kappa_3 \approx 1.089401427.$$

3 Valid all- n max-min lower bounds

For $0 < m < 1$, put

$$t = t_m, \quad L = \log(1/t).$$

Independently offer each bidder a price having density

$$g(p) = \frac{\mathbf{1}_{[t,1]}(p)}{pL},$$

in a random order, until the item sells. Define

$$a = \Pr(V \geq P) = \mathbb{E} \left[\frac{\log(V/t)_+}{L} \right],$$

$$b = \mathbb{E}[P \mathbf{1}_{\{V \geq P\}}] = \frac{1}{L} \int_t^1 \Pr(V \geq p) dp.$$

Since

$$\int_t^1 \Pr(V \geq p) dp \geq m - t = tL,$$

we have $b \geq t$.

To bound a , consider

$$f_t(v) = \frac{\log(v/t)_+}{L}.$$

If $L \leq 1$, its least concave majorant is $v \mapsto v$. If $L \geq 1$, its least concave majorant is

$$\widehat{f}_t(v) = \begin{cases} v/(eL), & v \leq et, \\ \log(v/t)/L, & v \geq et. \end{cases} \quad (10)$$

The two pieces are tangent at et , and

$$\log(v/t) \leq \frac{v}{et} \quad (t \leq v \leq et).$$

Thus Jensen's inequality, together with $m = t(1 + L)$, gives

$$a \leq \alpha_m = \begin{cases} m, & 0 < L \leq 1, \\ \frac{1+L}{eL}, & 1 \leq L \leq e-1, \\ \frac{\log(1+L)}{L}, & L \geq e-1. \end{cases} \quad (11)$$

Writing

$$S_n(x) = \sum_{j=0}^{n-1} (1-x)^j,$$

the sequential mechanism earns

$$bS_n(a) \geq t_m S_n(\alpha_m) = B_n(m). \quad (12)$$

The log-uniform construction is not defined at the endpoints. Separately,

$$V_n(0) = 0, \quad V_n(1) = 1,$$

so one may set $B_n(0) = 0$ and $B_n(1) = 1$.

For the no-reserve second-price auction, let

$$h_n(q) = 1 - (1 - q)^n - nq(1 - q)^{n-1}, \quad p_n = \frac{1}{(n-1)^2},$$

and

$$\rho_n = (1 - p_n)^{n-2}(1 + (n-1)p_n).$$

The lower convex envelope of h_n is

$$\underline{h}_n(q) = \begin{cases} h_n(q), & q \leq p_n, \\ 1 - \rho_n(1 - q), & q \geq p_n. \end{cases} \quad (13)$$

Indeed,

$$h'_n(q) = n(n-1)q(1-q)^{n-2},$$

and the line in (13) is tangent at p_n and meets h_n again only at 1.

If $Q(x) = \Pr(V \geq x)$, then

$$\text{Rev}_F(\text{SPA}) = \int_0^1 h_n(Q(x)) dx.$$

Since $\int_0^1 Q(x) dx = m$, Jensen's inequality gives

$$\inf_{F \in \mathcal{F}_m} \text{Rev}_F(\text{SPA}) = \underline{h}_n(m). \quad (14)$$

Equality is attained by two-point laws: for $m \leq p_n$, by

$$(1 - m)\delta_0 + m\delta_1;$$

for $m \geq p_n$, by

$$(1 - p_n)\delta_{(m-p_n)/(1-p_n)} + p_n\delta_1.$$

Consequently,

$$\boxed{\max\{B_n(m), \underline{h}_n(m)\} \leq V_n(m) \leq U_n(m).} \quad (15)$$

As $m \downarrow 0$,

$$t_m \sim \frac{m}{\log(1/m)}, \quad \alpha_m \rightarrow 0,$$

so

$$\boxed{V_n(m) \sim \frac{nm}{\log(1/m)} \quad (m \downarrow 0).} \quad (16)$$

4 Corrected two-threshold pooling family

Fix $0 \leq b < c \leq 1$, and draw a random priority order before bids are submitted.

- If some bid is at least c , allocate to the earliest such bidder.
- Otherwise, if some bid is at least b , allocate to the earliest such bidder.
- Otherwise, do not sell.

The critical payments are explicit:

- when there is no high bidder, the winner pays b ;
- if there are at least two high bidders, the winner pays c ;
- if there is exactly one high bidder, that bidder pays b when it precedes every other b -eligible bidder, and otherwise pays c ;
- losers pay zero.

Thus the mechanism is universally dominant-strategy incentive compatible (DSIC) and ex-post individually rational (IR).

Let

$$r = \Pr(V \geq b), \quad q = \Pr(V \geq c),$$

and

$$J_n(q, r) = \begin{cases} \frac{(1-q)^n - (1-r)^n}{r-q}, & q < r, \\ n(1-q)^{n-1}, & q = r. \end{cases}$$

The probability of a b -payment is $rJ_n(q, r)$, while the probability of a c -payment is $K_n(q) - qJ_n(q, r)$. Hence

$$R_{b,c}(q, r) = cK_n(q) + (br - cq)J_n(q, r). \quad (17)$$

The corrected closure of feasible pairs is

$$\mathcal{C}_{b,c,m} = \left\{ (q, r) : \begin{array}{l} 0 \leq q \leq r \leq 1, \\ br + (c-b)q \leq m, \\ m \leq b + (c-b)r + (1-c)q, \\ b = 0 \implies r = 1, \\ m = c = 1 \implies q = 1 \end{array} \right\}. \quad (18)$$

The first two moment inequalities follow from the category masses

$$1-r, \quad r-q, \quad q$$

on $[0, b)$, $[b, c)$, and $[c, 1]$, respectively. Conversely, every point in (18) is a limit of exact-mean distributions with those tail probabilities. Therefore the correct finite-dimensional formula is

$$\boxed{\inf_{F \in \mathcal{F}_m} \text{Rev}_F(P_{b,c}) = \min_{(q,r) \in \mathcal{C}_{b,c,m}} R_{b,c}(q, r).} \quad (19)$$

For the verifier's counterexample

$$n = 3, \quad b = 0, \quad c = \frac{1}{5}, \quad m = \frac{3}{5},$$

the condition $b = 0$ forces $r = 1$, and the upper mean inequality forces $q \geq 1/2$. Hence

$$\inf R_{0,1/5} = \frac{1}{5} (1 - (1 - \frac{1}{2})^2) = \frac{3}{20}.$$

The spurious point $q = r = 3/5$ is no longer feasible. Likewise, when $m = c = 1$, (18) forces $q = r = 1$, as it must because $F = \delta_1$.

Strict improvement at $n = 3$, $m = 2/3$

For $n = 3$,

$$h_3(q) = 3q^2 - 2q^3$$

and

$$h_3(q) - \left(\frac{9}{8}q - \frac{1}{8} \right) = \frac{(1-q)(4q-1)^2}{8} \geq 0.$$

Thus the unique SPA worst-case law at mean $2/3$ is

$$F_0 = \frac{3}{4}\delta_{5/9} + \frac{1}{4}\delta_1, \quad \text{Rev}_{F_0}(\text{SPA}) = \frac{5}{8}.$$

For $b = 1/2$ and $c = 9/10$, this law gives $r = 1$, $q = 1/4$, and

$$\text{Rev}_{F_0}(P_{1/2,9/10}) = \frac{9}{10} \cdot \frac{37}{64} + \left(\frac{1}{2} - \frac{9}{40} \right) \frac{9}{16} = \frac{27}{40}.$$

This pointwise improvement implies a strict robust improvement. The exact-mean ambiguity set is weakly compact. SPA revenue is weakly continuous and has the unique minimizer F_0 . Pooling revenue is continuous at F_0 , since F_0 has no atom at either threshold. Therefore there exist a neighborhood $\mathcal{O}$ of F_0 , $\delta > 0$, and $\eta > 0$ such that

$$R_{\text{pool}} \geq R_{\text{SPA}} + \delta \quad \text{on } \mathcal{O}$$

and

$$R_{\text{SPA}} \geq \frac{5}{8} + \eta \quad \text{outside } \mathcal{O}.$$

For sufficiently small $\varepsilon > 0$, the mixture

$$(1 - \varepsilon) \text{SPA} + \varepsilon P_{1/2,9/10}$$

earns strictly more than $5/8$ everywhere. Hence

$$\boxed{\frac{5}{8} < V_3(2/3) \leq 1 - \frac{\kappa_3}{3} \approx 0.636866191.} \tag{20}$$

5 Correct saddle-candidate conditions

Suppose a mechanism M actually guarantees $U_n(m)$. Since

$$\text{Rev}_{F_m^\dagger}(M) \leq \text{OPT}_n(F_m^\dagger) = U_n(m),$$

it must be Bayesian-optimal for $F_m^\dagger$.

All resulting allocation statements hold only $F_m^{\dagger n}$ -almost surely.

- On the low branch, the top atom has ironed virtual value 1, while every lower type has ironed virtual value 0. Whenever a top type occurs, the allocation must go entirely to top types. Allocation on no-top profiles is not forced.
- On the high branch, top types have ironed virtual value 1, while lower types have value $a = 1 - \lambda$. Top types must be prioritized. If $m_n < m < 1$, then $a > 0$, so the item must be allocated with probability one on no-top profiles.
- At $m = m_n$, one has $a = 0$, so allocation on no-top profiles is not forced.
- The cases $m = 0$ and $m = 1$ are the separate degenerate laws δ_0 and δ_1 .

These Bayesian conditions are far from sufficient for robustness. On the low branch, offering price 1 to all bidders is optimal against $F_m^\dagger$, earning $K_n(t_m)$, but earns zero against δ_m .

On the high branch, let

$$s_m = a + \lambda\tau_n.$$

Sequentially offer the prices

$$(1, \dots, 1, s_m),$$

with $n - 1$ offers at 1 and a final offer at s_m . Against $F_m^\dagger$, the revenue is

$$1 - (1 - \tau_n)^{n-1} + s_m(1 - \tau_n)^{n-1} = 1 - \lambda(1 - \tau_n)^n = U_n(m).$$

But against δ_m this mechanism earns only $s_m < U_n(m)$. Thus optimality against the adversary-first law alone does not provide a saddle point.

There are also precise variational restrictions. Symmetrize M , let P_M be its bounded total-payment function, and define

$$\begin{aligned} a_M(v) &= \int P_M(v, v_2, \dots, v_n) d(F^\dagger)^{n-1}, \\ b_M(v, w) &= \int P_M(v, w, v_3, \dots, v_n) d(F^\dagger)^{n-2}. \end{aligned}$$

For a fixed finite signed measure H ,

$$\begin{aligned} R_M(F^\dagger + \varepsilon H) &= R_M(F^\dagger) + n\varepsilon \int a_M dH \\ &\quad + \binom{n}{2} \varepsilon^2 \iint b_M(v, w) dH(v) dH(w) + O_H(\varepsilon^3). \end{aligned} \tag{21}$$

The remainder is not uniform over unrestricted H .

Define

$$T_+(F^\dagger) = \left\{ H : \begin{array}{l} H([0, 1]) = 0, \int v dH(v) = 0, \\ F^\dagger + \varepsilon H \geq 0 \text{ for all sufficiently small } \varepsilon \geq 0 \end{array} \right\}$$

and

$$T_\pm(F^\dagger) = T_+(F^\dagger) \cap [-T_+(F^\dagger)].$$

If $F^\dagger$ minimizes R_M , then

$$\int a_M dH \geq 0 \quad (H \in T_+) \quad (22)$$

and

$$\int a_M dH = 0 \quad (H \in T_\pm), \quad (23)$$

while the second-order condition

$$\iint b_M dH dH \geq 0 \quad (24)$$

is necessary only when the first variation in the given feasible direction vanishes. In particular, (24) holds for two-sided directions, but need not hold for a one-sided direction with strictly positive first variation.

For example, take the mechanism posting price 0.8, and let

$$F = \delta_{0.2}, \quad G = 0.2\delta_1 + 0.8\delta_0, \quad H = G - F.$$

Then $F + \varepsilon H$ is a feasible one-sided mean-preserving path, and F is a global revenue minimizer. Nevertheless,

$$0.8[1 - (1 - 0.2\varepsilon)^n]$$

has a positive linear coefficient but a negative quadratic coefficient. This directly disproves the former unrestricted quadratic condition.

Finally, every $G \in \mathcal{F}_m$ gives the feasible one-sided direction $G - F^\dagger$. Hence a saddle candidate must satisfy the global first-order condition

$$\int a_M dG \geq \int a_M dF^\dagger \quad (G \in \mathcal{F}_m). \quad (25)$$

For $0 < m < 1$, elementary two-moment convex separation makes (25) equivalent to the existence of α, β such that

$$\boxed{a_M(v) \geq \alpha + \beta v \quad (0 \leq v \leq 1), \quad a_M(v) = \alpha + \beta v \quad F^\dagger\text{-a.e.}} \quad (26)$$

Indeed, the lower boundary of the closed convex hull of

$$\{(v, a_M(v)) : 0 \leq v \leq 1\}$$

has a supporting line at the mean m , and equality after integration forces pointwise equality $F^\dagger$ -almost everywhere. Thus (26) includes:

- equality throughout the continuous component almost everywhere;
- equality at the positive atom at 1;
- the global off-support inequalities, including values below the support.

This condition is stronger than the former continuous-support affine condition, but remains only a necessary first-order condition.

6 Exact remaining gap

The repaired results do not identify an unrestricted optimal mechanism, its actual worst-case distribution, or the exact value of $V_n(m)$ for general $n \geq 3$.

A mechanism satisfying

$$\text{Rev}_F(M) \geq U_n(m) \quad (F \in \mathcal{F}_m) \quad (27)$$

would prove $V_n(m) = D_n(m)$, but this is only one possible way the problem could end. Failure or nonexistence of such a mechanism would merely rule out an upper-bound-attaining pure saddle; it would not determine the lower value $V_n(m)$, an optimal mechanism below $D_n(m)$, or its least-favorable distributions.

Thus the strongest established conclusions are the exact adversary-first formula, the all- n bounds and the small-mean asymptotic, the corrected pooling reduction, and

$$\frac{5}{8} < V_3(2/3) \leq 0.636866191.$$

Closing the unrestricted problem requires either constructing a mechanism satisfying (27), or finding a lower value together with both a mechanism attaining it and a matching adversarial upper certificate.

Remark. For values in $[0, \bar{v}]$, substitute $m = \mu/\bar{v}$ and multiply all prices and revenues by $\bar{v}$.

References

- [1] Nir Bachrach, Inbal Talgam-Cohen. *Distributional Robustness: From Pricing to Auctions*. Economics and Computation (EC), 2022. <https://doi.org/10.48550/arxiv.2205.09008>